

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning,

and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I K. Reshma/ 192565062 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Reshma

ARTICLE REVIEW

TASK 1 – ARTICLE SUMMARY

1(a) Citation Details and Problem

The selected research focuses on the application of **Deep Reinforcement Learning (DRL)** to autonomous vehicle control. The main objective is to enable an autonomous vehicle to make appropriate driving decisions by interacting with a changing traffic environment.

Domain/Application Area:

Autonomous driving and intelligent transportation systems.

Specific ML Problem:

The vehicle needs to select suitable actions such as acceleration, deceleration, lane changing, and maintaining a safe distance from other vehicles. These decisions must consider safety, traffic efficiency, and driving comfort.

The problem can be represented using reinforcement learning concepts:

- **State:** Current vehicle position, speed, surrounding vehicles, traffic conditions.
- **Action:** Acceleration, braking, lane change, etc.
- **Reward:** Safe and efficient driving.
- **Environment:** Dynamic road and traffic conditions.

1(b) Article Objective and Research Gap

The main objective of the research is to develop an intelligent decision-making system for autonomous vehicles using reinforcement learning. Traditional rule-based vehicle-control systems require manually designed rules for different traffic situations. Such rules may work in simple environments but can become difficult to maintain when the environment becomes complex and unpredictable.

The research identifies the need for an adaptive learning mechanism that can learn suitable driving decisions through interaction with the environment. Reinforcement learning provides such a mechanism because an agent learns by receiving rewards or penalties for its actions. The research therefore proposes a deep reinforcement learning approach that can automatically learn driving policies from traffic conditions.

The proposed approach attempts to improve important driving objectives such as **safety, efficiency, and comfort**. Instead of depending completely on predefined rules, the vehicle learns which actions produce better long-term outcomes. This can make autonomous driving systems more flexible in changing traffic conditions.

The research gap is mainly related to developing intelligent decision-making methods that can handle the complexity of real-world traffic. The proposed reinforcement-learning model attempts to address this gap by learning an appropriate policy through repeated interaction with the environment.

TASK 2 – METHODOLOGY ANALYSIS

2(a) Machine Learning Techniques

The main technique used in the research is **Deep Reinforcement Learning**.

Reinforcement learning consists of four major components:

1. **Agent** – The autonomous vehicle.
2. **Environment** – The road and surrounding traffic.
3. **Action** – Driving decisions taken by the vehicle.
4. **Reward** – Feedback received after performing an action.

The agent observes the current state and selects an action. The environment then changes according to that action and provides a reward. The process continues until the agent learns a good driving policy.

Dataset / Input Information

The system uses traffic and vehicle-state information such as:

- Vehicle speed
- Vehicle position
- Relative distance from other vehicles
- Surrounding traffic
- Lane information
- Acceleration

- Road conditions
- Vehicle movement

The data/environment is used to train and evaluate the reinforcement-learning agent. Unlike conventional supervised learning, the agent does not simply learn from labelled input-output pairs. Instead, it learns through interaction and rewards.

Basic Working

Traffic Environment → State Observation → RL Agent → Action → Environment → Reward → Policy Update

The model repeatedly performs this process until it learns a suitable driving policy.

2(b) Mapping to CO4

The technique directly maps to the **Reinforcement Learning** component of CO4.

CO4 Topic	Connection with Article
Inductive Learning	The system learns useful behaviour from experience
Decision Trees	Not the primary technique
Statistical Learning	Not the primary technique
Reinforcement Learning Main technique used for autonomous decision-making	

Methodological Strength

A major strength is that reinforcement learning can **adapt its decisions according to environmental feedback**. This is useful for autonomous vehicles because traffic conditions continuously change.

Methodological Limitation

One limitation is that reinforcement-learning systems generally require extensive training and simulation. A policy that performs well in simulation may not always perform equally well in unpredictable real-world traffic.

TASK 3 – RESULTS AND CRITICAL EVALUATION

3(a) Key Results

The research evaluates the learned driving policy using performance measures related to safety, efficiency, and driving behaviour.

Typical evaluation parameters include:

- Collision/safety rate
- Average speed
- Travel efficiency
- Acceleration
- Driving comfort
- Cumulative reward
- Convergence of the learned policy

Comparative Evaluation

Method	Safety	Efficiency	Adaptability
Rule-Based Method	Good in predefined situations	Moderate	Low
Traditional Control	Good	Moderate	Moderate
Reinforcement Learning	High	High	High

The reinforcement-learning approach is advantageous because it can learn a policy based on experience rather than relying completely on predefined rules.

The assignment specifically asks students to report metrics such as accuracy, F1-score, AUC-ROC, or reward convergence **as presented by the selected article**.

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Therefore, exact numerical values should be copied directly from the article's results section rather than estimated.

3(b) Critical Evaluation

The results demonstrate that reinforcement learning can be useful for autonomous decision-making. However, the results should be interpreted carefully.

Possible Dataset Bias

If the training environment contains only a limited range of traffic situations, the learned policy may not generalize to all real-world conditions.

For example, the model may perform well on:

- Normal traffic
- Standard road layouts
- Moderate traffic density

but may perform differently under:

- Heavy traffic
- Unexpected pedestrian movement
- Accidents
- Poor weather
- Unusual driver behaviour

Evaluation Bias

Performance obtained in simulation does not necessarily guarantee identical performance on real roads. Real-world environments contain many unpredictable factors.

Additional Experiments

The research could be strengthened by:

1. Testing on multiple traffic datasets.
2. Testing under different weather conditions.
3. Comparing several reinforcement-learning algorithms.
4. Performing real-world or hardware-in-the-loop testing.
5. Testing the system with different traffic densities.
6. Conducting long-term safety evaluation.

These experiments would provide stronger evidence of generalization and reliability.

3(c) Real-World Applicability

The technique can potentially be deployed in:

- Autonomous cars
- Smart transportation systems
- Connected vehicles
- Intelligent traffic management
- Highway driving systems
- Vehicle platooning
- Robotic transportation

Industry Challenges

1. Data availability:

Large and diverse traffic data is required to train reliable models.

2. Computational requirements:

Deep reinforcement-learning models can require considerable computational resources.

3. Safety:

An incorrect decision by an autonomous vehicle can result in accidents.

4. Generalization:

A model trained in one city or traffic environment may not work equally well in another.

5. Ethical issues:

Autonomous systems must make safe decisions when faced with unavoidable risks.

6. Real-time performance:

The system must make decisions quickly enough for practical vehicle operation.

TASK 4 – REFLECTION AND LEARNING OUTCOME

4(a) Three Key Insights

Insight 1 – Learning Through Rewards

I learned that reinforcement learning is different from supervised learning because the agent learns through **rewards and penalties**.

CO4 Connection: Reinforcement Learning.

The autonomous vehicle receives feedback based on its driving behaviour and gradually improves its policy.

Insight 2 – State, Action and Environment

The article helped me understand the relationship between **state, action, reward, and environment**.

For an autonomous vehicle:

- State = Traffic and vehicle information
- Action = Driving decision
- Reward = Safety and efficiency feedback
- Environment = Road and traffic

CO4 Connection: Reinforcement-learning framework.

Insight 3 – Importance of Generalization

I learned that achieving good performance during training does not necessarily mean that an AI system will perform well in every real-world situation.

CO4 Connection: Machine-learning model evaluation and decision-making.

A reinforcement-learning system should be tested under different environments to verify its reliability.

4(b) Proposed Extension

If I were to extend this research, I would propose a **hybrid Deep Reinforcement Learning system using real-world traffic datasets and simulated environments**.

The model could first be trained in simulation because simulation allows thousands of driving scenarios to be generated safely. The trained model could then be evaluated using real-world traffic data.

The experiment could compare:

Existing DRL → Improved DRL → Hybrid DRL

The improved system could include additional environmental information such as:

- Weather
- Pedestrian movement
- Traffic density
- Road conditions
- Emergency vehicles

Feasibility

This experiment is feasible because traffic simulators can generate large numbers of training scenarios without putting real people or vehicles at risk.

Expected Impact

The proposed enhancement could improve:

- Safety
- Generalization
- Decision-making
- Robustness

- Real-world applicability

It would also provide a stronger connection between reinforcement-learning research and practical autonomous-driving applications.

CONCLUSION

The reviewed research demonstrates the importance of **Reinforcement Learning in intelligent decision-making**. Autonomous vehicles operate in dynamic environments where predefined rules may not be sufficient for every situation. Reinforcement learning provides an adaptive approach in which an agent learns suitable actions from environmental feedback.

The article helped develop a better understanding of CO4 concepts, particularly the relationship between **state, action, reward, environment, and policy learning**. Although reinforcement learning provides promising results, issues such as training requirements, safety, generalization, computational cost, and real-world validation must be addressed before large-scale deployment.

Overall, reinforcement learning has significant potential for autonomous vehicles and other intelligent decision-making applications.